

TWINKIT · DESIGN DECISIONS REGISTER

TwinKit design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 3 on 2026-10-02 (TWK-DDR-003 accepted, US915 first build, first candidate partner); moved to decisions made
0.1	2026-10-02	Amish Chadha	Register opened with the open decisions, purchase checks and value engineering from the constructable design (TWK-DDR-003)

TwinKit design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , TWK-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The 9-module enclosure: whether it has moulded bosses for the computer, moulded vents, a flat front wall where the 20 mm and 16.2 mm holes go, and how its cover is held	Sets whether the floor holes and vent slots are cut or already there (build plan section 3.3)	TWK-DDR-003, P2, P3, P5
2	The concentrator HAT passes the 40-pin header through, or names free GPIO and ground pins, and sits on the computer's 58 x 49 hole pattern	The power-fail pair connects there; the 16 mm standoffs and header extender assume that pattern and height	TWK-DDR-003, P3, P9
3	The active cooler is no taller than 8 mm above the board	It must clear the HAT on 16 mm standoffs	TWK-DDR-003, P3
4	The UPS module: 3 modules wide, no internal battery, a LiFePO4 setting at 14.6 V and 0.5 A or less, a 0 to 45 °C charge window, and a power-fail output that is a dry contact or open collector	Width sets the rail length; the output type decides whether the pair can go straight to a GPIO pin (safety stop S3)	<code>bom/bom.csv</code> line 11; TWK-DDR-003, P1, P9
5	The backup pack is no larger than 38 x 70 x 38 mm with a lead of about 300 mm and its own BMS and fuse	It must fit the 3-module battery box with 3.5 mm to the gland nut	<code>bom/bom.csv</code> line 12; TWK-DDR-003, P1
6	The DC-DC converter is 2 modules wide	Sets the rail length (345 mm of 350 mm used with end stops)	<code>bom/bom.csv</code> line 10
7	The computer runs from the converter's 5.1 V through its USB-C socket without power negotiation	It then limits its USB sockets to low power, which the gateway does not use; if it refuses to boot, power it through its header pins instead	TWK-DDR-003, P9
8	The network coupler's panel hole (20 mm assumed) and the rail's factory slot positions	Hole sizes in build plan sections 3.2 and 3.3	TWK-DDR-003, P5, P6

■ Value engineering

Value-engineering target: USD 300 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 334 (USD 34 over the target). Main cost drivers and savings worth trying:

- The largest lines are the concentrator HAT (USD 80), the computer (USD 65), the UPS module (USD 30), the backup pack (USD 25), the gateway enclosure and the DC-DC converter (USD 18 each) and the aluminium plate (USD 15).
- Making the design constructable added USD 44 against the concept's USD 290: the battery box (USD 8), end stops (USD 3), board mounting kit (USD 5), cable entries (USD 10), fixings and wiring (USD 9) and the aluminium plate in place of plywood (USD 9 more).
- Savings worth trying: a 9 mm plywood plate with threaded inserts in place of tapped aluminium (about USD 6 saved); an enclosure that already has bosses and vents for the computer (saves the standoff kit, about USD 3, and the cutting time); a DIN UPS module with a built-in LiFePO4 pack, if one meeting line 11 exists, which would remove the battery box (about USD 8); and a single supplier for the computer, cooler, card and standoffs. None of these changes what the gateway does.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 items D1 to D7: 8-channel LoRaWAN concentrator, 4 GB Pi 5 class computer, TimescaleDB, lightweight open-source stack with a TwinKit twin service, FieldNode battery and solar charge as the first example twin, the YAML twin file fields, an off-the-shelf LiFePO4 pack with built-in BMS	Amish: "i accept all your recommendations, go with them across all repos."	TWK-DDR-001, TWK-DDR-002
2026-09-25	Budget raised from USD 200 to USD 300 to keep LoRaWAN compatibility (O1)	Amish, same instruction	TWK-DDR-002
2026-09-25	R1 judged with adaptive data rate required (N1); R12 defined at light load with heavy jobs in cool hours, a larger or metal enclosure as the fallback (N2)	Amish, same instruction	TWK-DDR-002
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a spending limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	<code>.kit/STANDARDS.md</code> section 18
2026-10-02	Design for construction accepted: the changes P1 to P9 of TWK-DDR-003, as made	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	TWK-DDR-003
2026-10-02	Region and radio band of the first build: US915, with the US915 concentrator HAT and a 915 MHz antenna, to match FieldNode's decided first variant	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	<code>bom/bom.csv</code> line 7; FND-DDR-001, O1
2026-10-02	Co-design partner outside the lab: the first candidate to approach for the operator view is a small municipal water utility in the Dallas and Fort Worth area, found through the Texas Water Utilities Association, with Dallas Makerspace as the build and test partner	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	TWK-DDR-002, O2